Precalculus

Նախամաթանալիզ

COURSE CODE	MTH-PC
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Typically Grade 11 or 12
DURATION	108–126 instructional hours (approx. 90 lessons)
PREREQUISITE	Algebra II (MTH-A2), or Algebra II with Trigonometry, or placement by assessment.
LEADS TO	Calculus (MTH-CA), AP Calculus AB (MTH-APCA) or BC (MTH-APCB)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Precalculus prepares a student to succeed in Calculus, which means two things: complete fluency with the function families and their algebra, and a first acquaintance with the ideas of limit and rate of change that Calculus will formalize. The course covers functions and their transformations in depth, polynomial and rational functions, exponential and logarithmic functions, trigonometry through identities and polar form, systems and matrices, conic sections, parametric equations, sequences and series, and an introduction to limits and the derivative concept.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Students fail Calculus for algebraic reasons far more often than for conceptual ones. Վարժարան therefore treats Precalculus as an algebra course with a calculus horizon: the algebraic manipulations that Calculus takes for granted — factoring, rational expression manipulation, exponent and logarithm laws, trigonometric simplification, domain analysis — are practiced to genuine automaticity, because a student who must think about the algebra has no attention left for the calculus. At the same time, the final unit introduces limits and the derivative honestly rather than as a preview, so that the first weeks of Calculus are recognition rather than shock.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Factors and manipulates rational expressions fluently	The strongest single predictor of Calculus performance; remediate aggressively if weak.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Applies exponent and logarithm properties without hesitation	Required continuously from Unit 3 onward.
Knows the unit circle and can produce exact values without reference	Prerequisite for Units 4 and 5; a very common gap.
Analyzes a function's domain, range, and transformations	The organizing framework of the entire course.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Analyze functions using domain, range, symmetry, transformations, composition, and inverses.
2. Analyze polynomial and rational functions completely, including end behaviour, zeros, and asymptotes.
3. Model with exponential and logarithmic functions and solve the associated equations.
4. Use the unit circle and radian measure to evaluate and analyze trigonometric functions.
5. Graph and model with trigonometric functions and verify trigonometric identities.
6. Solve trigonometric equations and apply the Laws of Sines and Cosines.
7. Represent complex numbers in polar form and apply De Moivre's theorem.
8. Solve systems of equations and inequalities, including nonlinear systems, using matrices where appropriate.
9. Perform matrix operations and use matrices to solve systems and to represent transformations.
10. Analyze conic sections from their equations and their geometric definitions.
11. Work with parametric and polar representations of curves and convert among forms.
12. Work with vectors in two and three dimensions, including the dot and cross products.
13. Analyze sequences and series, including the binomial theorem and mathematical induction.
14. Evaluate limits informally and numerically, and interpret the derivative as an instantaneous rate of change.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 12-13 LESSONS

Functions and Their Analysis

Ֆունկցիաներ և դրանց վերլուծություն

TOPICS COVERED

- Function definition, notation, domain, and range in interval notation
- The parent function catalogue extended
- Transformations and their order of application
- Symmetry: even, odd, and neither
- Increasing, decreasing, and constant intervals
- Relative and absolute extrema
- Average rate of change and the difference quotient
- Piecewise functions, step functions, and continuity described informally
- Operations on functions: sum, difference, product, quotient and their domains
- Composition of functions and decomposition of a composite
- One-to-one functions and the horizontal line test
- Inverse functions, their domains, and verification by composition
- Modelling with function operations and compositions

LEARNING OBJECTIVES

1. State domain and range in interval notation for any function in the catalogue.
2. Compute and simplify the difference quotient for polynomial and rational functions.
3. Decompose a composite function into components in more than one way.
4. Find and verify an inverse function, restricting the domain where necessary.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- States domain and range correctly in interval notation in 9 of 10 trials.
- Simplifies the difference quotient at 88% accuracy.
- Finds, restricts, and verifies inverse functions in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The difference quotient is not simplified, leaving h in the denominator.	Simplification is the point — it is what makes the limit computable. This is the direct precursor of the derivative definition, and it should be named as such.

MISCONCEPTION	RECOMMENDED RESPONSE
Interval notation and inequality notation are mixed, or brackets and parentheses are interchanged.	Require interval notation exclusively from the first lesson. Precision here prevents a great deal of Calculus confusion.
An inverse is claimed without restricting the domain of a non-one-to-one function.	Check one-to-one first, always. The restriction is part of the answer.

TEACHING NOTE

The difference quotient introduced here is the derivative in all but name. Say so explicitly. Students who have simplified difference quotients for a term meet the derivative definition as something familiar.

UNIT 2 · 11-12 LESSONS

Polynomial and Rational Functions

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TOPICS COVERED

- Quadratic functions in all three forms and their applications
- Polynomial end behaviour, zeros, and multiplicity
- Synthetic and long division; the Remainder and Factor theorems
- The Rational Root Theorem and complete factorization
- The Fundamental Theorem of Algebra and conjugate pairs
- Complete polynomial graph analysis and sketching
- Polynomial inequalities and sign analysis
- Rational function domain, holes, and vertical asymptotes
- Horizontal and slant asymptotes; end behaviour of rational functions
- Complete rational function graph analysis
- Rational inequalities
- Partial fraction decomposition
- Modelling with polynomial and rational functions

LEARNING OBJECTIVES

1. Produce a complete graph analysis of a polynomial or rational function.
2. Determine every asymptote and hole with justification.
3. Solve polynomial and rational inequalities using sign analysis.
4. Decompose a rational expression into partial fractions.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces complete graph analyses correctly in 4 of 5 tasks.
- Solves polynomial and rational inequalities at 88% accuracy.
- Performs partial fraction decomposition correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A graph is assumed never to cross a horizontal asymptote.	It may cross in the middle; the asymptote describes end behaviour only. Produce an example and graph it.
Sign analysis intervals are determined without testing.	Test a point in each interval. Sign does not necessarily alternate — even multiplicity does not change sign.
A slant asymptote is sought when the degree difference is not exactly one.	Compare degrees first, then choose the case. Make the degree comparison a written first step.

TEACHING NOTE

Partial fractions is included specifically because integration by partial fractions appears in Calculus BC. Teaching it here rather than mid-Calculus removes an algebraic obstacle at a moment when the student should be attending to the calculus.

UNIT 3 · 9-10 LESSONS

Exponential and Logarithmic Functions

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TOPICS COVERED

- Exponential functions, transformations, and asymptotic behaviour
- The number e and its origin in continuous growth
- Logarithmic functions as inverses; domain and asymptotes
- Properties of logarithms and the change of base formula
- Solving exponential and logarithmic equations with domain checking
- Exponential and logarithmic inequalities
- Compound and continuous interest; effective annual rate
- Growth, decay, half-life, and doubling time
- Newton's law of cooling; logistic growth models
- Linearizing exponential and power data with logarithms
- Regression with exponential and logarithmic models
- Logarithmic scales in science and engineering

LEARNING OBJECTIVES

1. Solve exponential and logarithmic equations and inequalities with full domain analysis.
2. Build and interpret an exponential, logarithmic, or logistic model of a real situation.
3. Linearize data using logarithms and interpret the resulting linear fit.
4. Explain the origin of e as a limit of compound interest.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves exponential and logarithmic equations at 90% accuracy with domain checking.
- Builds and interprets models correctly in 4 of 5 tasks.
- Linearizes data and interprets the fit in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Logarithm properties are applied to sums or to the argument incorrectly.	Test numerically. A student who has produced their own counterexample does not repeat the error.
Domain restrictions are neglected when solving logarithmic equations.	Every candidate solution must be checked against the domain of the original equation, not the simplified one.
e is regarded as an arbitrary constant with no meaning.	Derive it as the limit of $(1 + 1/n)^n$. This also gives the student an early, concrete encounter with a limit.

TEACHING NOTE

Introducing e as a limit here is a deliberate early exposure to limiting processes. It costs one lesson and makes the formal treatment in Unit 8 substantially less abstract.

UNIT 4 · 13-14 LESSONS

Trigonometric Functions and Identities

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TOPICS COVERED

- Angles, radian measure, arc length, and angular velocity
- The unit circle and exact values in all quadrants
- Right triangle trigonometry and applications
- Graphs of all six functions with complete transformation analysis
- Modelling periodic phenomena and fitting sinusoids to data
- Inverse trigonometric functions, restricted domains, and compositions
- Fundamental identities and systematic verification strategies
- Sum, difference, double-angle, half-angle, and power-reducing formulas
- Product-to-sum and sum-to-product formulas
- Solving trigonometric equations over an interval and in general
- Trigonometric equations requiring identities or factoring
- The Law of Sines with the ambiguous case; the Law of Cosines
- Triangle area formulas including Heron's formula
- Applied triangle problems in surveying and navigation

LEARNING OBJECTIVES

1. Produce exact values and graph any transformed trigonometric function without reference material.
2. Verify identities using a named systematic strategy.
3. Solve trigonometric equations completely, including general solutions.
4. Solve any triangle and resolve the ambiguous case.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces exact values without reference in 9 of 10 trials.

- Verifies identities of moderate difficulty in 4 of 5 tasks.
- Solves trigonometric equations completely at 88% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The power-reducing and half-angle formulas are confused with each other.	They are algebraically related. Derive one from the other rather than storing both separately.
Solutions are lost by dividing an equation by a trigonometric factor.	Factor instead of dividing, always. Dividing discards solutions and the loss is silent.
Calculator dependence hides an inability to produce exact values.	Non-calculator work appears in every assignment and every assessment in this unit, without exception.

TEACHING NOTE

This unit is the longest in the course and should be. Trigonometric fluency is required continuously in Calculus, and a student who is slow here will be slow everywhere in Calculus.

UNIT 5 · 9–10 LESSONS

Systems, Matrices, and Linear Programming

Համակարգեր, մատրիցներ և գծային ծրագրավորում

TOPICS COVERED

- Systems of linear equations in two and three variables
- Gaussian elimination and row echelon form
- Matrix notation, equality, and the augmented matrix
- Matrix addition, scalar multiplication, and matrix multiplication
- The identity matrix and the inverse of a matrix
- Determinants of 2 × 2 and 3 × 3 matrices; Cramer's rule
- Solving systems with matrix inverses
- Matrices as transformations of the plane
- Nonlinear systems of equations
- Systems of inequalities and feasible regions
- Linear programming and optimization at the vertices
- Applications: resource allocation, mixture, and network problems

LEARNING OBJECTIVES

1. Solve a three-variable system by Gaussian elimination and by matrix inverse.
2. Perform all matrix operations and compute determinants and inverses.
3. Solve a nonlinear system algebraically and confirm graphically.
4. Formulate and solve a linear programming problem with a stated objective function.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves three-variable systems at 90% accuracy by at least two methods.
- Performs matrix operations including inverses at 90% accuracy.
- Formulates and solves a linear programming problem correctly in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Matrix multiplication is assumed commutative.	AB and BA are generally different and may not both exist. Compute a counterexample in the first lesson on multiplication.
Matrix dimensions are not checked before multiplying.	Inner dimensions must agree. Write the dimensions above each matrix before attempting the product.
A nonlinear system is assumed to have a single solution.	It may have zero, one, two, or more. Graph first to determine how many solutions to expect.

TEACHING NOTE

Presenting matrices as transformations of the plane, rather than only as a solution device, gives the topic conceptual weight and connects it to the transformational geometry the student already knows.

UNIT 6 · 10-11 LESSONS

Conics, Parametric Equations, and Polar Coordinates

Կոնական հատույթներ, պարամետրական հավասարումներ և բևեռային կոորդինատներ

TOPICS COVERED

- The conic sections as plane sections of a cone
- The parabola: focus, directrix, and the standard equations
- The ellipse: foci, axes, eccentricity, and the standard equations
- The hyperbola: foci, asymptotes, and the standard equations
- Completing the square to identify a conic from a general equation
- The discriminant test for conic type
- Applications: orbits, reflectors, and navigation systems
- Parametric equations and the parameter as time
- Eliminating the parameter to find a rectangular equation
- Parametric equations of lines, circles, and projectile motion
- Polar coordinates and multiple representations
- Converting between polar and rectangular form for points and equations
- Graphing polar equations and testing for symmetry
- Polar equations of the conic sections

LEARNING OBJECTIVES

1. Identify a conic from its general equation and produce its complete analysis.
2. Derive a conic equation from its geometric focus-directrix definition.
3. Eliminate the parameter from a parametric pair and describe the orientation of the curve.
4. Graph a polar equation and identify its symmetry.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Identifies and analyzes conics from general equations in 9 of 10 trials.
- Converts between parametric and rectangular forms at 88% accuracy.
- Graphs polar equations correctly in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Ellipse and hyperbola equations are confused because only the sign differs.	Sum of distances versus difference of distances. Return to the geometric definitions, which are genuinely different.
Orientation and domain are lost when the parameter is eliminated.	The parametric curve carries direction and may be only part of the rectangular curve. State the restriction explicitly.
Polar graphs are plotted point by point without using symmetry.	Test for symmetry first; it typically reduces the work by a factor of two or four.

TEACHING NOTE

Deriving at least one conic equation from its focus-directrix definition is worth the time. It shows the student that these equations are consequences of geometric conditions rather than arbitrary forms to be recognized.

UNIT 7 · 10-11 LESSONS

Vectors, Sequences, and Series

Վեկտորներ, հաջորդականություններ և շարքեր

TOPICS COVERED

- Vectors in two dimensions: magnitude, direction, and components
- Vector operations and unit vectors
- The dot product, the angle between vectors, and projection
- Vectors in three dimensions and the coordinate space
- The cross product and its geometric meaning
- Applications: work, torque, force resolution, and navigation
- Sequences: explicit and recursive definitions
- Arithmetic and geometric sequences and series
- Sigma notation and its manipulation
- Infinite geometric series and convergence
- The binomial theorem and Pascal's triangle
- Mathematical induction and the structure of an inductive proof
- Counting: permutations, combinations, and the multiplication principle
- Probability applications of counting techniques

LEARNING OBJECTIVES

1. Compute dot and cross products and interpret each geometrically.
2. Determine whether an infinite geometric series converges and find its sum.

3 Expand a binomial power using the binomial theorem.
4 Construct a complete proof by mathematical induction.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Performs vector operations in two and three dimensions at 90% accuracy.
- Works with sequences and series including sigma notation at 90% accuracy.
- Produces a valid induction proof in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The dot and cross products are confused: the dot product expected to be a vector.	The dot product is a scalar; the cross product is a vector perpendicular to both. State the output type before computing.
The inductive step is treated as an assumption of the result itself.	The step assumes the case n = k and proves n = k + 1. Write both statements explicitly before beginning.
A sequence is confused with a series.	A sequence is a list; a series is a sum. Distinguish them in every problem statement.

TEACHING NOTE

Mathematical induction is many students' first genuinely formal proof technique after Geometry, and it is the most common one in university mathematics. Require complete, well-structured write-ups rather than sketches.

UNIT 8 · 9-10 LESSONS

Introduction to Limits and the Derivative

Ներածություն սահմաններին և ածանցյալին

TOPICS COVERED

- The intuitive idea of a limit
- Estimating limits numerically and graphically
- One-sided limits and when a limit fails to exist
- Algebraic evaluation of limits: substitution, factoring, rationalizing, and common denominators
- Limits involving infinity and their relationship to asymptotes
- Continuity at a point and on an interval
- Classifying discontinuities: removable, jump, and infinite
- The Intermediate Value Theorem stated and applied
- The tangent line problem and the slope of a curve
- The difference quotient and the definition of the derivative
- The derivative as an instantaneous rate of change
- Computing a derivative from the definition for simple functions
- The derivative function and its graphical relationship to the original
- The area problem and an informal introduction to accumulation

LEARNING OBJECTIVES

1. Evaluate a limit algebraically using an appropriate technique and justify the choice.
2. Determine continuity at a point and classify any discontinuity.
3. Compute a derivative from the definition for a polynomial or simple rational function.
4. Interpret the derivative as an instantaneous rate of change with units in an applied context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Evaluates limits algebraically at 88% accuracy across all techniques.
- Classifies discontinuities correctly in 9 of 10 trials.
- Computes derivatives from the definition correctly in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The limit at a point is assumed to equal the function value there.	They agree exactly when the function is continuous there. Use a removable discontinuity to separate the two ideas permanently.
0/0 is treated as an answer rather than as an indication that more work is required.	It is an indeterminate form, which means the limit exists or not depending on the function. Factor, rationalize, or combine and try again.
The derivative is described as a slope without saying of what.	It is the slope of the tangent line at a point, and equivalently an instantaneous rate of change. Require both descriptions with units in applied problems.

TEACHING NOTE

This unit is deliberately substantial rather than a preview. Students who arrive at Calculus having already computed derivatives from the definition spend the first month consolidating rather than struggling, and the difference in confidence is marked.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Algebra II retention weighted toward rational expressions, logarithms, and the unit circle.
Unit Check	End of each unit · 60 min	Unit mastery criteria with a required non-calculator section.
Algebra Fluency Probe	Every 2nd lesson · 5 min	Factoring, rational expressions, exponent and logarithm laws, and exact trigonometric values — the skills Calculus assumes.
Modelling Task	Once per unit from Unit 3	Open-ended problems requiring model selection, construction, and defence.

ASSESSMENT	WHEN	WHAT IT MEASURES
Mid-Year Assessment	After Unit 4	Cumulative Units 1–4; weighted toward functions and trigonometry.
End-of-Course Assessment	After Unit 8	All course outcomes; produces a formal Calculus readiness recommendation, including an AP AB or BC recommendation where relevant.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard Year	2 lessons/week, 60–75 min	Full course in approximately 45 weeks.
Accelerated	3 lessons/week	Full course in approximately 30 weeks.
Support / Co-Seated	2 lessons/week alongside school	Paced to the student's school syllabus, with Վարժարան supplying diagnosis and Calculus-readiness work.
Calculus Bridge	3 lessons/week, 10 weeks	Units 1, 2, and 8 — for students entering Calculus with algebraic gaps.
Summer Program	4 lessons/week, 8 weeks	Units 4 and 8 — trigonometry and limits.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Sixty to seventy-five minutes per lesson: fifteen to twenty problems in three parts — current content, cumulative algebra review, and one extended or modelling problem. The cumulative section is the most important part of the assignment in this course and is never optional: it maintains the algebraic automaticity that Calculus will assume. Every assignment includes non-calculator work. Domain analysis and solution checking are required throughout, and answers to applied problems must carry units and an interpretive sentence.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Analyzes any function completely: domain, range, symmetry, transformations, inverse.
- Simplifies the difference quotient fluently.
- Produces complete graph analyses of polynomial and rational functions.
- Performs partial fraction decomposition.
- Solves exponential and logarithmic equations with full domain analysis.
- Produces exact trigonometric values and graphs without reference material.
- Verifies identities using named systematic strategies.
- Solves trigonometric equations completely, including general solutions.
- Solves systems by elimination, matrices, and inverses; formulates linear programs.
- Analyzes all conic sections and works with parametric and polar forms.
- Computes dot and cross products and applies vectors to physical problems.
- Works with sequences, series, the binomial theorem, and mathematical induction.
- Evaluates limits algebraically and classifies discontinuities.
- Computes derivatives from the definition and interprets them as rates of change.
- Completed Precalculus — formally assessed as ready for Calculus.

MATERIALS AND RESOURCES

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- Graphing calculator (TI-84 family or equivalent) — required, with a stated policy on when it may not be used.
- Dynamic graphing software for conics, polar curves, and parametric motion.
- Computer algebra system for checking, used only after work is completed by hand.
- Real data sets for the modelling tasks.
- Վարժարան Precalculus problem bank — approximately 2,400 items including a 400-item non-calculator Calculus-readiness sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.